

1. Familiarize yourself with the basics of Assembly Language:

Before diving into NASM, MASM or Emu8086, it's important to understand the fundamentals of Assembly Language.

Start by learning the basic concepts such as registers, memory, instructions, and the syntax used to write Assembly code.

2. Choose a program to install and analyze: MASM, NASM, or EMU8086. If none of the three programs work for you, find an application that you can use to work in assembly language. Download and install the version that is compatible with your system.

3. Write simple programs: Start by writing simple programs in application you choose to get a feel for the language. Start with basic programs like printing messages on the screen, reading input from the user, and performing arithmetic operations. This will help you understand how the program works and get comfortable with the syntax.

4. Debug your programs: Debugging is an essential part of programming.

Learn how to use debugging tools to identify and fix errors in your code.

This will help you become more efficient in your programming and also give you a better understanding of how your code works.

You have to go through all these steps and confirm the completion by text or screenshot, and of **course to write 3 programs** in application with the theme of your choice and comment each line of code it executes.

The report must contain:

1. Introduction

- Brief explanation of Assembly Language
- Importance of low-level programming

2. Purpose of the report

Chosen application (MASM, NASM, or EMU8086) and why it was selected

3. Familiarization with Assembly Basics

4. Installation and Setup

- Description of the selected program
- Steps of installation
- System compatibility (OS, version)
- Initial configuration
- Screenshot: installed program / working environment

5. Simple Programs

- Write and explain at least 3 programs exercises:

For each program:

- a) Source code
- b) Line-by-line comments (explanation of each instruction)
- c) Output description
- d) Screenshot: code + result

Conclusion